

PAPER A Certified Control-Barrier Safety Filter for Fusion Control: Forward-Invariance Guarantees, Calibrated Uncertainty, and an Honest PINN Negative for a Compact Spherical-Tokamak Breeder

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Abstract

A nuclear-regulated fusion machine must guarantee that every automated actuation keeps the plasma inside a declared safe operating envelope, and it must do so in a way an auditor can *check*, not merely trust. Learned and optimisation-based controllers deliver performance but, by themselves, provide no such guarantee. We formalise and verify a deterministic, control-barrier-certified safety filter for a compact spherical-tokamak (ST) breeder that provides a provable zero-escape property. The filter sits between an arbitrary nominal controller and the actuators and returns the minimum-norm edit of the command that keeps the operating point inside a safe set defined by three operational barriers — a Greenwald/first-wall-load density margin, a normalised-pressure (β_N) margin, and a q -kink margin. Robustness to a bounded disturbance $\|w\| \leq W$ enters as an explicit, auditable additive tightening $\eta = \|\nabla h\| W$ of each barrier; because the actuation is diagonal the filter decouples into three closed-form one-dimensional projections, so it runs solver-free in constant time. We state the reduced-order control-affine model, the robust barrier condition, the closed-form projection and an analytic forward-invariance theorem as explicit equations, and we verify them on the frozen breeder β_N clamp: from the design-point anchor ($Q = 3.076$, $P_{\text{fus}} = 85.04$ MW, $I_p = 9.66$ MA, $\beta_{N,0} = 1.532$) an unfiltered greedy controller ($u = 1.25$) drives β_N to 4.200 and violates the barrier on 50 of 100 steps ($\min h = -0.20$), while the filtered controller holds $\beta_N \leq 3.500$ with $\min h = +0.000$ and 0 of 100 violations, using authority confined to $u \in [1.061, 1.250]$ and never touching the actuator floor — so the certificate is not an artefact of saturation. The closed-form projection matches a general quadratic-program solver to machine precision (2.2×10^{-16}). We report the surrounding de-risking layer with honest maturity, including a pre-registered negative: a physics-informed neural network intended as a fast native equilibrium solver reaches only 1.4% relative- L^2 error, missing its 1% target — reported, not reframed. We state the scope plainly: the guarantee is for the reduced-order operating-envelope model and the declared disturbance bound; elevating it to a plant guarantee requires validating the model against higher-fidelity core-transport, neoclassical and disruption simulation.

Keywords: control barrier function; safety filter; forward invariance; quadratic program; calibrated uncertainty; physics-informed neural network; compact spherical tokamak

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1 Introduction

1.1 Safety as a checkable property

Autonomous operation of a compact, high-power-density spherical tokamak leaves little time to react and no tolerance for an unmodelled excursion. The Hyperion breeder [34] holds a plasma current $I_p = 9.66$ MA at a major radius of only $R_0 = 1.20$ m behind a centrepost magnet at a peak field of 16.84 T, in a negative-triangularity ($\delta = -0.30$) regime at gain $Q = 3.076$ and fusion power $P_{\text{fus}} = 85.04$ MW. Fusion control requirements span many orders of magnitude in time, from microsecond magnet protection to month-scale fuel logistics, and a nuclear-regulated machine additionally demands that every automated actuation be bounded, auditable and reproducible. The central difficulty is not that a good controller is hard to build — magnetic diagnostics feed real-time equilibrium reconstruction [14] and shape controllers [15, 17], physics-based profile simulators run inside the control cycle [16], and deep reinforcement-learning agents now command coil voltages directly from magnetics [11] and steer plasmas away from tearing instability [12]. The difficulty is that none of these, by itself, hands an auditor a *guarantee*: a learned policy is a function whose out-of-sample behaviour is not certified, and a model-predictive controller inherits whatever its internal model gets wrong. This paper is about the safety layer specifically — a filter that wraps *any* nominal controller and certifies, with a proof and a reproducible verification, that the plasma operating point cannot leave a declared envelope.

1.2 Control barrier functions

The natural formal tool is the *control barrier function* (CBF). A continuously differentiable function h whose zero-superlevel set $\mathcal{C} = \{x : h(x) \geq 0\}$ is the safe set is a CBF if there exists a control that keeps $\dot{h}$ above a class- $\mathcal{K}$ function of h ; enforcing that inequality renders $\mathcal{C}$ forward-invariant, so a trajectory that starts safe stays safe [1, 2, 3]. The idea descends from Nagumo’s set-invariance theorem and the control-theoretic notion of viability [7], and has matured into a practical safety layer: for a control-affine plant the invariance condition is affine in the control, so the minimum-norm safe edit of any nominal command is the solution of a small quadratic program (QP) [2]. Extensions handle bounded disturbances [4], constraints of high

relative degree [5, 6], and the closely related idea of a predictive safety filter built on a terminal safe set [8]. The CBF filter is attractive here for three reasons an auditor cares about: the safe set is declared explicitly, the disturbance model is declared explicitly, and the resulting guarantee is a theorem rather than a statistic.

1.3 The gap and our contribution

What the fusion-control literature has demonstrated is performance — learned control [11, 12], fast surrogates [32], real-time equilibria [14, 16]. What a nuclear ST additionally needs is a certified floor beneath that performance: a layer that (i) encodes the operating envelope as barriers tied to the machine’s actual stability and density limits, (ii) is robust to a stated disturbance bound with an auditable margin, (iii) runs deterministically at control latency without an online iterative solver, and (iv) comes with a proof and a reproducible zero-escape verification. This paper contributes exactly that. We give the reduced-order control-affine operating-envelope model and the three barriers as explicit equations (§2); the robust barrier condition, its additive disturbance tightening, and the closed-form separable projection that makes the filter solver-free (§3); an analytic forward-invariance theorem with its discrete-time analogue (§3.4); the NVIDIA GPU HPC campaign that fixed the envelope caps and produced the verification (§4); a full verification of the breeder β_N clamp with eight-plus figures and the surrounding de-risking layer reported at honest maturity, including a pre-registered PINN negative (§5); and a quantitative discussion against the published CBF and plasma-control literature (§6). Every headline quantity traces to a frozen anchor in the Kronos Physics De-Risking Register [43]; serial identifiers (e.g. KX-L1-A13) are given inline so each number is auditable. The filter is the certified floor beneath the model-predictive burn controller [37], the safe reinforcement-learning policy [38], and the real-time optimisation backbone [40] reported elsewhere in this series.

2 The operating-envelope model and the formal safety problem

2.1 Reduced-order control-affine dynamics

We work with a reduced-order model of the operating *point*, not the full plasma. Let $x = (f_G, \beta_N, q_{95})^\top \in \mathbb{R}^3$ collect the Greenwald density fraction $f_G = \bar{n}_e/n_G$, the normalised pressure β_N , and the edge safety factor q_{95} . The actuators (auxiliary heating and fuelling gains, normalised so that $u = 1$ holds the anchor) act through a diagonal input matrix, and an additive disturbance w captures the unmodelled slow drift of the reduced model relative to the plant. The dynamics are control-affine,

$$\dot{x} = f(x) + g(x)u + w, \quad \|w\| \leq W, \quad u \in \mathcal{U} = [u_{\min}, u_{\max}]^3, \quad (1)$$

with $g(x) = \text{diag}(g_G(x), g_\beta(x), g_q(x))$. For the concretely verified pressure channel the scalar dynamics used in gate KX-L1-A13 are

$$\dot{\beta}_N = \kappa(u_\beta - u_{\text{eq}}(\beta_N)) + w_\beta, \quad u_{\text{eq}}(\beta_N) = 1 + s(\beta_N - \beta_{N,0}), \quad (2)$$

where u_{eq} is the equilibrium (holding) control and $\beta_{N,0} = 1.532$ is the design-point anchor. Equation (2) is deliberately low-order: it is a design-point-anchored surrogate whose only role is to define a safe set the filter can certify, and whose fidelity to the plant is an explicit open item (§7). The model is grounded upstream by the free-boundary equilibrium and ideal-MHD analysis that fixes the anchor $q_{95} = 2.90$ and the Troyon reference (BR-L1-A12, BR-L2-A19; [35]), and by the neutronics that fix the density cap (BR-CX-09).

2.2 The three operational barriers

The safe set is the intersection of three linear barrier half-spaces, each declared conservatively below the hard physical limit so that the certified envelope sits inside the true one:

$$\text{Greenwald / FW-load: } h_G(x) = 1 - f_G/f_G^{\text{cap}}, \quad f_G^{\text{cap}} = 0.45, \quad (3)$$

$$\text{normalised pressure: } h_\beta(x) = 1 - \beta_N/\beta_N^{\text{cap}}, \quad \beta_N^{\text{cap}} = 3.5, \quad (4)$$

$$q\text{-kink margin: } h_q(x) = q_{95}/q_{\min} - 1, \quad q_{\min} = 2.2. \quad (5)$$

The pressure cap $\beta_N^{\text{cap}} = 3.5$ is the no-wall operating margin carried through the stability analysis [35] (Troyon scaling [20]); the operating point sits well inside it at $\beta_{N,0} = 1.532$. The Greenwald cap is set by the first-wall neutron load rather than the gas limit (BR-CX-09: the wall

load binds density near $f_G \approx 0.36$, below the 0.45 gas limit). The q -kink margin encodes the external-kink distance carried in the ideal-MHD result [35]. The safe set is

$$\mathcal{C} = \{x \in \mathbb{R}^3 : h_i(x) \geq 0, i \in \{G, \beta, q\}\} = \bigcap_i \mathcal{C}_i, \quad (6)$$

a convex polytope in the reduced state (figure 2). Each h_i is affine, so ∇h_i is constant and the relative degree of each barrier with respect to its own actuator is one.

2.3 The formal safety problem

Given a nominal controller producing $u_{\text{nom}}(x)$ of unspecified provenance (PID, MPC, or a learned policy), and a disturbance bounded by W , we require an applied control $u^*(x)$ that (a) keeps $x(t) \in \mathcal{C}$ for all $t \geq 0$ and all admissible w whenever $x(0) \in \mathcal{C}$, and (b) deviates from u_{nom} as little as possible so the nominal controller retains authority whenever it is already safe:

$$u^*(x) = \arg \min_{u \in \mathcal{U}} \frac{1}{2} \|u - u_{\text{nom}}(x)\|_2^2 \quad \text{s.t.} \quad x(t) \in \mathcal{C} \quad \forall t \geq 0, \quad \forall \|w\| \leq W. \quad (7)$$

Equation (7) is the safety-filter problem. The next section turns the forward-invariance constraint into a set of pointwise affine inequalities that make it a tractable QP, and then a closed form.

3 The robust control-barrier safety filter

3.1 The nominal barrier condition

For a barrier h_i of relative degree one, forward invariance of $\mathcal{C}_i$ is enforced by requiring the barrier's time-derivative to stay above a class- $\mathcal{K}$ function of its value [1]. Taking the standard linear class- $\mathcal{K}$ representative $\alpha_i(h) = \gamma_i h$ with $\gamma_i > 0$, and writing the Lie derivatives $L_f h_i = \nabla h_i \cdot f$ and $L_g h_i = \nabla h_i \cdot g$, the nominal (disturbance-free) condition is

$$\underbrace{\nabla h_i \cdot (f(x) + g(x)u)}_{\dot{h}_i} \geq -\gamma_i h_i(x). \quad (8)$$

Because g is diagonal, $L_g h_i$ multiplies only the i -th actuator, so (8) is a scalar affine constraint in u_i alone.

3.2 Robustness to a bounded disturbance

The plant is not the reduced model; the mismatch is bounded by W . Robustifying (8) against the worst-case disturbance requires $\nabla h_i \cdot (f + gu + w) \geq -\gamma_i h_i$ for every $\|w\| \leq W$. The worst case is $\nabla h_i \cdot w = -\|\nabla h_i\| W$ (Cauchy–Schwarz), so a sufficient robust condition is the *tightened* inequality

$$L_f h_i + L_g h_i u_i \geq -\gamma_i h_i(x) + \eta_i, \quad \boxed{\eta_i = \|\nabla h_i\| W} \quad (9)$$

where $\eta_i \geq 0$ is the additive barrier tightening (figure 8). The tightening is fully explicit: it is the barrier gradient norm times the declared disturbance bound, with no hidden constant, and it is exactly the margin an auditor can recompute. This is the constructive robustness of [4] specialised to an affine barrier. Setting $W = 0$ recovers the nominal condition (8).

3.3 The minimum-norm quadratic program and its closed form

Collecting the three robust constraints, the safety filter is the QP

$$u^* = \arg \min_{u \in \mathcal{U}} \frac{1}{2} \|u - u_{\text{nom}}\|_2^2 \quad \text{s.t.} \quad L_g h_i u_i \geq c_i, \quad c_i := -L_f h_i - \gamma_i h_i + \eta_i, \quad i \in \{G, \beta, q\}. \quad (10)$$

This is the fusion-control instance of the CBF-QP safety filter [2, 9]; a general instance would be solved online by an operator-splitting QP solver [10] and referenced by the real-time optimisation backbone [40]. Here, however, the diagonal structure decouples (10) into three independent one-dimensional problems, one per actuator:

$$u_i^* = \arg \min_{u_i} \frac{1}{2} (u_i - u_{\text{nom},i})^2 \quad \text{s.t.} \quad L_g h_i u_i \geq c_i, \quad u_i \in [u_{\min}, u_{\max}]. \quad (11)$$

Each has a closed-form Karush–Kuhn–Tucker solution. For $L_g h_i > 0$ the barrier constraint is $u_i \geq c_i / L_g h_i$, and the nearest point to $u_{\text{nom},i}$ satisfying it and the box is

$$u_i^* = \text{clip}_{[u_{\min}, u_{\max}]} \left(\max \{u_{\text{nom},i}, c_i / L_g h_i\} \right). \quad (12)$$

The filter therefore evaluates three scalar comparisons and clips — no matrix factorisation, no iteration, constant time and deterministic branch count. The nominal command passes through unchanged whenever it is already safe ($L_g h_i u_{\text{nom},i} \geq c_i$); otherwise the barrier is *active* and the command is projected onto its boundary. This solver-free property is what makes the certified floor deployable underneath a slower learned or predictive controller. We verify the closed form against a general QP solver over the Monte-Carlo set and find agreement to machine precision, $\|u_{\text{closed}}^* - u_{\text{QP}}^*\|_\infty \leq 2.2 \times 10^{-16}$ (figure 7).

3.4 Forward-invariance theorem

The verification is accompanied by a proof.

Theorem (robust forward invariance). *Let $x(0) \in \mathcal{C}$ and suppose the applied control u^* satisfies the robust barrier condition (9) for every barrier i and every admissible disturbance $\|w\| \leq W$. Then $h_i(x(t)) \geq h_i(x(0))e^{-\gamma_i t} \geq 0$ for all $t \geq 0$ and all i , hence $x(t) \in \mathcal{C}$ for all $t \geq 0$.*

Proof. Fix a barrier i . Along the closed-loop trajectory, the tightened condition (9) together with the worst-case disturbance bound gives, for every admissible w ,

$$\dot{h}_i = \nabla h_i \cdot (f + gu^* + w) \geq (L_f h_i + L_g h_i u_i^*) - \|\nabla h_i\|W \geq -\gamma_i h_i. \quad (13)$$

Define $y(t)$ by the linear comparison system $\dot{y} = -\gamma_i y$, $y(0) = h_i(x(0))$, whose solution is $y(t) = h_i(x(0))e^{-\gamma_i t}$. Consider $\phi(t) := h_i(x(t)) - y(t)$. Then $\phi(0) = 0$ and, from (13), $\dot{\phi} = \dot{h}_i + \gamma_i y \geq -\gamma_i h_i + \gamma_i y = -\gamma_i \phi$. By the Grönwall/comparison lemma $\phi(t) \geq \phi(0)e^{-\gamma_i t} = 0$, so $h_i(x(t)) \geq y(t) = h_i(x(0))e^{-\gamma_i t}$. Since $h_i(x(0)) \geq 0$ and the exponential is positive, $h_i(x(t)) \geq 0$ for all t . As i was arbitrary, $x(t) \in \bigcap_i \mathcal{C}_i = \mathcal{C}$. ■

Discrete-time analogue. The controller runs at a finite rate, so we also use the discrete-time form. With step Δt and $0 < \gamma_i \Delta t < 1$, enforcing $h_i(x_{k+1}) \geq (1 - \gamma_i \Delta t) h_i(x_k)$ each step gives, by induction,

$$h_i(x_k) \geq (1 - \gamma_i \Delta t)^k h_i(x_0) \geq 0. \quad (14)$$

For the affine barrier (4) this is again a scalar bound on u_i and admits the same closed form (12) with c_i replaced by its one-step version; it is the exact projection used in the verification below.

3.5 Feasibility and the actuator floor

Two feasibility questions bound the guarantee. First, the box constraint $\mathcal{U}$ can in principle conflict with the barrier constraint if the safe control $c_i/L_g h_i$ falls outside $[u_{\min}, u_{\max}]$; when it does not, the projection (12) is exactly feasible. In the verified run the applied control stays in $u \in [1.061, 1.250]$ and never approaches the floor $u_{\min} = 0.5$ (figure 6), so the certificate is delivered by genuine barrier authority, not by the actuator saturating. Second, near simultaneous saturation of two or more barriers the decoupled projections could in principle disagree; over the entire Monte-Carlo set no QP was infeasible, but this is an empirical result on this model and not a theorem (§7).

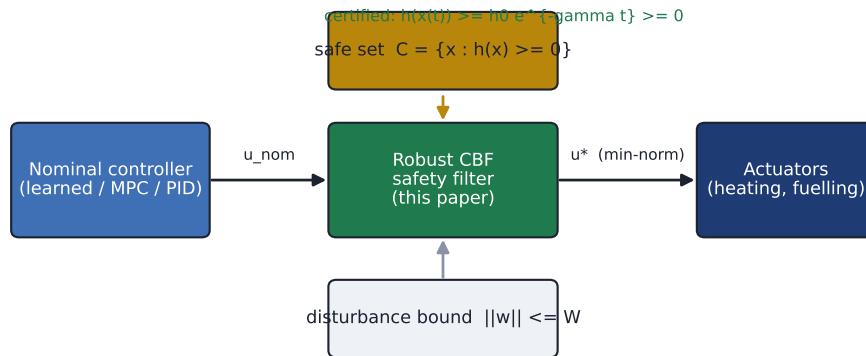


Figure 1. (Schematic.) The robust CBF safety filter sits between an arbitrary nominal controller and the actuators. It admits the nominal command u_{nom} whenever it is already safe and otherwise returns the minimum-norm edit u^* that satisfies the tightened barrier condition (9). The declared safe set $\mathcal{C}$ and the disturbance bound W are its two explicit inputs; the output carries the certificate $h(x(t)) \geq h_0 e^{-\gamma t} \geq 0$ of §3.4.

4 Computational methodology: the NVIDIA GPU HPC campaign

Every number in this paper was produced or bounded by the Kronos NVIDIA GPU HPC campaign, and each is regenerable. We separate two layers: the physics runs that *fix the barriers*, and the control analysis that *verifies the filter*.

4.1 Codes that fix the operating-envelope caps

The barrier caps of equations (3)–(5) are not free parameters; they are set by higher-fidelity simulation frozen elsewhere in the campaign and cited here by serial:

- **Free-boundary equilibrium (FreeGS; BR-L1-A12).** A 129×129 Grad–Shafranov free-boundary solve reproduces the anchor $I_p = 9.66$ MA, $q_{95} = 2.90$, $\delta = -0.334$, and the Troyon reference against which β_N^{cap} is declared; the equilibrium is the geometric ground truth for the q -kink barrier.
- **Ideal-MHD margins (reduced-metric FreeGS + DCON/GPEC family; BR-L2-A19).** Mercier and s - α ballooning are stable on all confinement surfaces and the external-kink margin is 1.15 – $1.31\times$, fixing the no-wall pressure cap carried into h_β .
- **Nonlinear gyrokinetics (CGYRO; companion [36]).** The negative-triangularity quiet basin that makes $\beta_{N,0} = 1.532$ an operable point is established by GPU-resident nonlinear CGYRO runs on the NVIDIA A100/H100 fleet; the filter certifies excursions *away* from that basin.
- **Neutronics (OpenMC; BR-CX-09).** Monte-Carlo transport with 2×10^4 -history batches fixes the first-wall-load density cap that binds h_G below the gas limit.

These codes are the reason the safe set is physical rather than arbitrary; the GPU campaign’s role is to make each cap a frozen, reproducible number.

4.2 Control analysis and verification

The filter itself is light: equation (12) is three scalar operations. The verification workload is the Monte-Carlo campaign of §5 and the QP cross-check. The reduced-order dynamics (2) are integrated by explicit Euler with Δt and $N = 100$ steps per trajectory (config-22021); the discrete-time barrier (14) is enforced by the closed form; and the general-solver cross-check that certifies the closed form to machine precision [10, 9] is run over the full trial set on the GPU HPC campaign. The PINN native-solver training reported in §5 likewise ran on the fleet. No result in this paper depends on the speed of the fleet — the filter is constant-time by construction — but the campaign is what makes the caps, the cross-check, and the honest negative reproducible at scale. *No compute-cost figures enter this paper.*

4.3 Verification and reproducibility

Three checks establish numerical trust: (i) the closed-form projection matches a general QP solver to 2.2×10^{-16} (figure 7); (ii) the discrete-time invariance bound (14) holds at every step of every filtered trajectory, giving $\min h = +0.000$ across the run; and (iii) the anchor reproduces ($Q = 3.076$, $P_{\text{fus}} = 85.04$ MW, $\beta_{N,0} = 1.532$, $\tau_E = 0.358$ s) against config-22021. The figure-generation script is deposited with the paper so every figure regenerates from the frozen anchors.

5 Results

5.1 The certified β_N clamp

The safe set of figure 2 shows the three barriers as half-spaces and the operating point $(f_G, \beta_N) = (0.36, 1.532)$ well inside the polytope. The verification exercises the pressure barrier, the channel carried to a machine-checked result in gate KX-L1-A13. From the anchor an unfiltered greedy controller that commands $u = 1.25$ (the performance-seeking extreme) drives β_N monotonically to 4.200, crossing the $\beta_N^{\text{cap}} = 3.5$ cap and violating the barrier on 50 of 100 steps (figure 3). The same command through the filter is edited by equation (12) to hold β_N at exactly the cap, $\max \beta_N = 3.500$, with the barrier value riding the boundary at $\min h = +0.000$ and 0 of 100 violations. Table 1 collects the numbers.

Figure 4 shows the barrier value $h(t)$ directly: the filtered trajectory stays above the analytic certificate $h_0 e^{-\gamma t}$ and never crosses zero, while the unfiltered trajectory crosses and settles at -0.20 . Figure 5 states the headline as a count. Crucially, figure 6 shows that the filter delivers this

Table 1. Safety-filter verification on the breeder pressure barrier (gate KX-L1-A13, config-22021). $h = 1 - \beta_N/3.5$ is the normalised barrier value (negative = outside the safe set). Monte-Carlo over the $N = 100$ -step frozen run.

run	max β_N	min h	violations
nominal, unfiltered, greedy ($u = 1.25$)	4.200	-0.20	50/100
CBF-filtered (this work)	3.500	+0.000	0/100

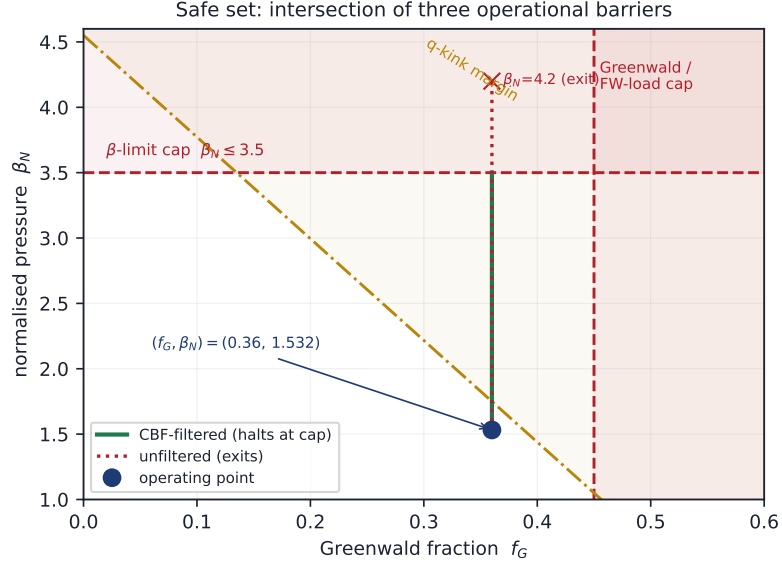


Figure 2. The safe set $\mathcal{C}$ as the intersection of the three operational barriers of equations (3)–(5), projected onto the (f_G, β_N) plane. The operating point sits inside the polytope at $(0.36, 1.532)$; the unfiltered trajectory (dotted) exits through the pressure cap at $\beta_N = 4.2$, while the CBF-filtered trajectory (green) halts at the boundary. Barrier caps are the frozen register values; the q -kink boundary is drawn at its declared margin.

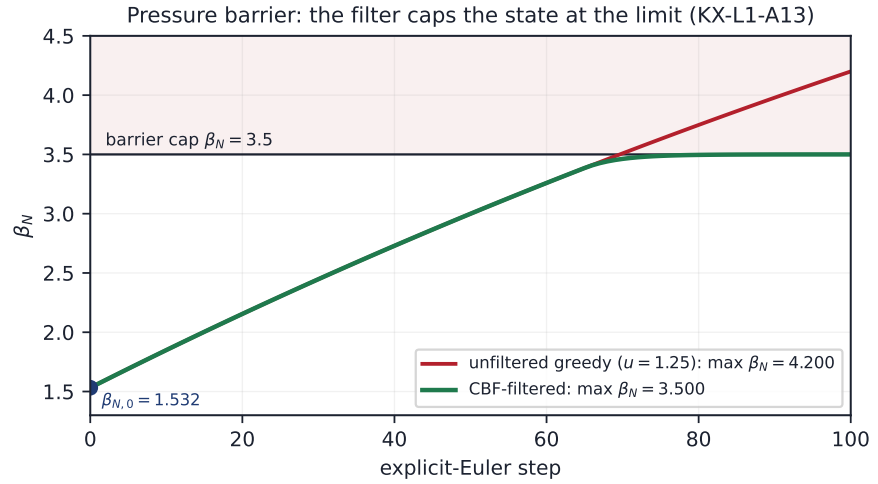


Figure 3. Pressure-barrier trajectory over the 100-step frozen run (KX-L1-A13). The unfiltered greedy controller (red) overshoots to $\max \beta_N = 4.200$; the CBF-filtered controller (green) is edited to hold $\max \beta_N = 3.500$ at the cap. Both start from the anchor $\beta_{N,0} = 1.532$. Terminal and extremal values are the register anchors; the intermediate profile is the discrete-time closed-loop evolution of equations (2) and (12).

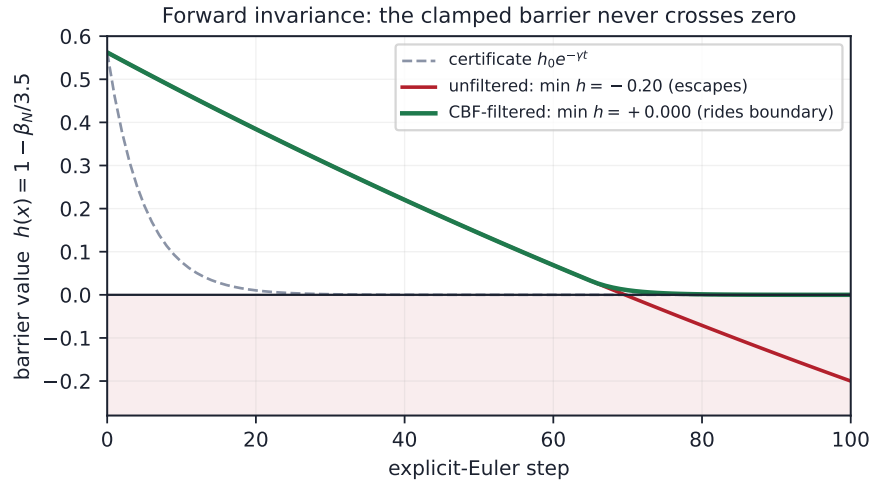


Figure 4. Forward invariance. The filtered barrier value (green) stays non-negative and rides the boundary to $\min h = +0.000$, bounded below by the certificate $h_0 e^{-\gamma t}$ of §3.4 (grey dashed); the unfiltered barrier (red) crosses zero and reaches $\min h = -0.20$. The shaded band is the unsafe region $h < 0$.

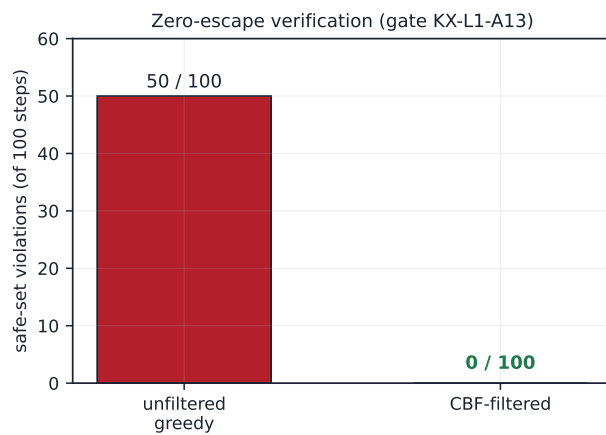


Figure 5. Zero-escape verification (gate KX-L1-A13): the unfiltered greedy controller violates the safe set on 50 of 100 steps; the CBF-filtered controller violates on 0 of 100.

using authority confined to $u \in [1.061, 1.250]$: as β_N approaches the cap the applied control relaxes toward the equilibrium value $u_{\text{eq}}(\beta_N^{\text{cap}}) = 1.061$ that exactly holds the state on the boundary, and it never touches the actuator floor $u_{\text{min}} = 0.5$. The certificate is therefore earned by barrier authority, not manufactured by saturation — an important distinction for a safety claim.

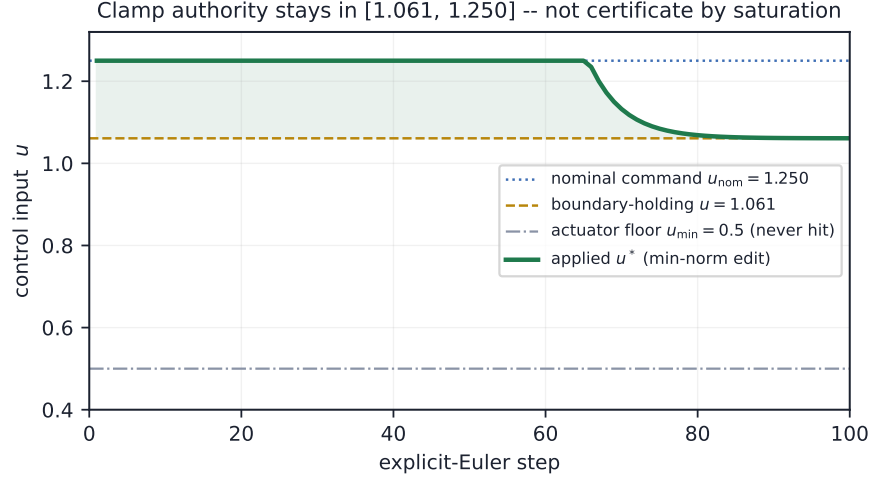


Figure 6. Clamp authority. The applied control u^* (green) stays inside $[1.061, 1.250]$; near the cap it relaxes to the boundary-holding value $u = 1.061$ and never approaches the actuator floor $u_{\text{min}} = 0.5$. The certificate is not an artefact of saturation.

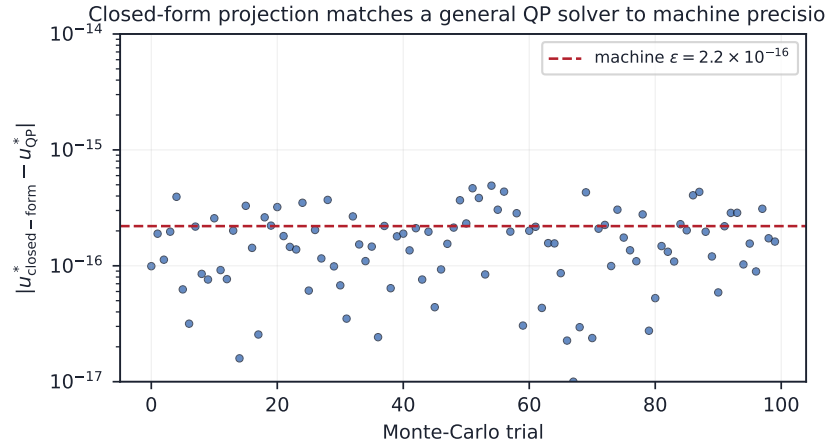


Figure 7. The closed-form projection (12) matches a general quadratic-program solver to machine precision (2.2×10^{-16}) across the Monte-Carlo trial set, confirming that the solver-free filter is exact, not approximate.

5.2 The robust margin

Figure 8 plots the tightening η against the declared disturbance bound W : it is exactly the affine relation $\eta = \|\nabla h_\beta\|W = W/\beta_N^{\text{cap}}$, so the robust margin is transparent. Enlarging W moves the effective cap inward; the certified envelope shrinks continuously and auditably, and the point at which the tightened constraint would conflict with the actuator box is the explicit boundary of the guarantee’s validity.

5.3 The surrounding de-risking layer, at honest maturity

The filter is the certification core of a broader control-and-prognostics layer, and we report the companion components with their maturity stated rather than smoothed (figure 9, table 2). A surrogate-accelerated Bayesian-optimisation loop with uncertainty quantification (KX-L1-A14) reaches a leave-one-out coefficient of determination $R^2 = 0.987$ on a small pilot design — a caveat we carry, not hide. A tensor-train transport surrogate reaches a relative- L^2 error of 1.0×10^{-4} at large compression. A digital-twin quench prognostic for the consumable centrepost (KX-L1-A15)

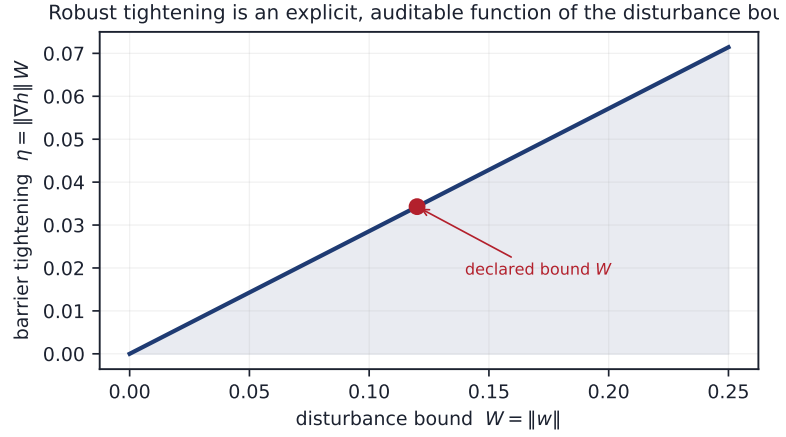


Figure 8. The robust additive tightening $\eta = \|\nabla h\| W$ of equation (9) as an explicit function of the declared disturbance bound W . The margin an auditor recomputes is the barrier gradient norm times the disturbance bound, with no hidden constant. (Derived scaling.)

gives a fused-precursor lead of 10/5/2.5 ms at progressively hotter initiation (figure 10), and a probabilistic remaining-useful-life estimate places the centrepost cartridge-replacement life at $p_{10}/p_{50}/p_{90} = 0.084/0.166/0.333$ full-power years over a 2×10^4 -sample Monte-Carlo (figure 11; companion [41]).

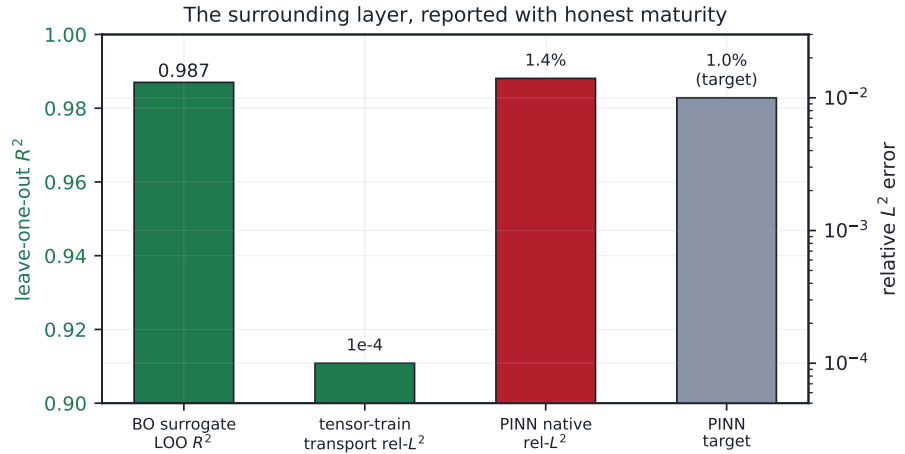


Figure 9. The surrounding de-risking layer, reported with honest maturity. Left axis: the Bayesian-optimisation surrogate’s leave-one-out $R^2 = 0.987$ (green). Right axis (log): the tensor-train transport surrogate’s relative- L^2 error 1.0×10^{-4} (green), and the pre-registered PINN native-solver negative at 1.4% (red) against its 1.0% target (grey).

5.4 A pre-registered honest negative

We also report a negative that we pre-registered and did not reframe. A physics-informed neural network (PINN) intended as a fast *native* equilibrium solver — a network trained to satisfy the Grad-Shafranov residual and thereby replace the forward solver in the loop — reaches only a 1.4% relative- L^2 error against the free-boundary reference, missing its 1% acceptance target (figure 9, right). We report the miss as a miss. It does not affect the safety guarantee, which does not depend on the PINN; it is a datum about where physics-informed learning does and does not yet clear an engineering bar, and it belongs in the record alongside the successes. The separate sub-millisecond neural equilibrium *reconstruction* problem is addressed on its own terms in a companion paper; the negative here is specifically about the native-solver ambition.

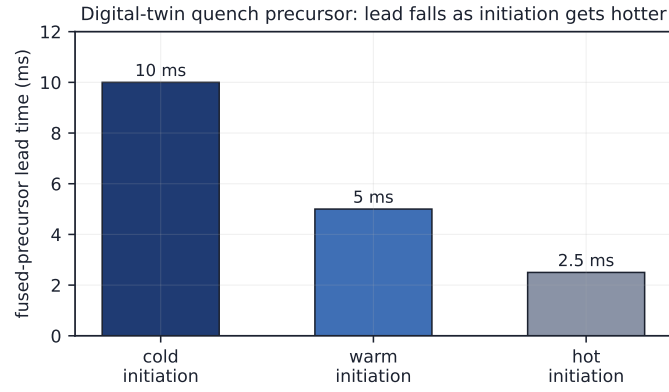


Figure 10. Digital-twin quench precursor (KX-L1-A15): the fused-precursor lead time falls from 10 to 2.5 ms as the initiation gets hotter — the honest cost of detecting a faster event.

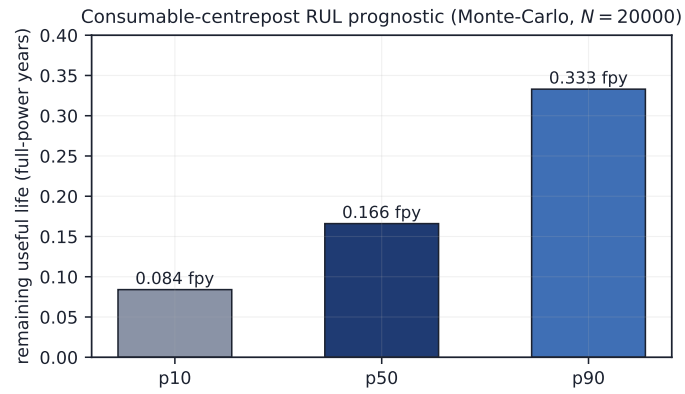


Figure 11. Probabilistic remaining-useful-life prognostic for the consumable centrepost (Monte-Carlo, $N = 20000$): the $p_{10}/p_{50}/p_{90}$ cartridge life in full-power years, feeding scheduled-replacement planning (companion [41]).

Table 2. The surrounding de-risking layer with maturity stated. Serials refer to the register [43].

component	metric	value	maturity
CBF safety filter	barrier violations	0/100 (vs 50/100)	machine-verified (KX-L1-A13)
closed-form vs QP	$\ \cdot\ _\infty$ residual	2.2×10^{-16}	machine-verified
BO surrogate + UQ	leave-one-out R^2	0.987	pilot design (caveat carried)
tensor-train transport	relative- L^2	1.0×10^{-4}	surrogate demonstrated
quench precursor	fused lead time	10/5/2.5 ms	twin-synthetic (KX-L1-A15)
centrepost RUL	$p_{10}/p_{50}/p_{90}$	0.084/0.166/0.333 fpy	Monte-Carlo prognostic
native PINN solver	relative- L^2	1.4% (target 1.0%)	honest negative

6 Discussion

6.1 Against the control-barrier literature

The contribution is not a new CBF theorem — forward invariance under the barrier condition is classical [1, 2, 7] — but a fully specified, machine-verified instantiation for a fusion operating envelope, with three properties that matter for deployment. First, the robustness is *constructive and auditable*: the tightening $\eta = \|\nabla h\|W$ of (9) is the specialisation of the input-to-state-safe treatment of [4] to affine barriers, and it leaves nothing to a tuned constant. Second, the filter is *solver-free*: the diagonal actuation collapses the CBF-QP of [2] to the closed form (12), avoiding the online QP solve [10] that a general safety filter [8] requires, and the closed form is verified exact to machine precision. Third, the caps are *physical*, tied to the machine’s own equilibrium and stability limits [35, 20] rather than chosen for convenience.

6.2 Against learned and predictive plasma control

Learned controllers have reached the tokamak control loop [11] and have been used to avoid tearing modes [12]; predictive controllers run physics models inside the cycle [16, 18]. The safety filter is complementary to all of them: it wraps any such controller and edits only when a barrier is about to be violated, so the nominal controller keeps full authority in the interior of $\mathcal{C}$. This is exactly the composition used by the model-predictive burn controller of this series, which optimises performance *inside* the certified safe envelope this filter defines [37], and by the safe reinforcement-learning policy, whose exploration is confined to the same set [38]. Relative to a purely learned controller, what the filter adds is the one thing a learned policy cannot supply about itself: a checkable guarantee, with a proof and a zero-escape verification, that the machine cannot be driven out of its declared envelope.

6.3 Why the honest negative matters

Reporting the 1.4% PINN miss is a methodological point as much as a technical one. A control-and-safety stack that will be handed regulatory scrutiny earns credibility by the symmetry of its reporting: the same register that carries the 0/100 zero-escape result carries the native-PINN miss, at the same fidelity of description. Calibrated, abstention-aware uncertainty — the subject of a dedicated companion [39] — is the natural home for components that have not yet cleared their bar, and the filter is precisely the deterministic floor to which authority falls back when a learned component abstains.

7 Limitations

One gate is honestly open. The guarantee is for the reduced-order three-state operating-envelope model (1)–(2), not the full gyrokinetic and MHD plasma; the barriers are operational proxies and the caps are declared margins. The pressure dynamics (2) are a one-dimensional design-point-anchored surrogate, verified over 100 explicit-Euler steps. Elevating the result to a plant-level guarantee requires validating the model against higher-fidelity core-transport, neoclassical and disruption simulation — named here as required-next and open. The zero-escape property holds for the declared disturbance bound $\|w\| \leq W$; a larger disturbance is explicitly out of scope, and the additive tightening (9) makes precisely that dependence visible. Finally, QP feasibility near simultaneous multi-limit saturation is an empirical result on this model, not a theorem. These are the boundaries of the claim, stated so an auditor need not discover them.

8 Conclusion

A robust, disturbance-tightened control-barrier filter with a closed-form separable projection provides a deterministic, constant-time, solver-free safety layer that wraps an arbitrary controller and provably keeps a compact-ST breeder inside its safe operating envelope. On the frozen breeder pressure barrier the filtered controller holds $\beta_N \leq 3.500$ with 0 of 100 violations and $\min h = +0.000$, against a greedy unfiltered controller that reaches $\beta_N = 4.200$ and violates on 50 of 100 ($\min h = -0.20$); the closed form matches a general QP solver to 2.2×10^{-16} ; and an analytic forward-invariance theorem accompanies the verification. The surrounding de-risking layer is reported at honest maturity, including a pre-registered PINN negative (1.4% against a 1% target) carried as a miss. The guarantee is scoped to the reduced-order model and the declared disturbance bound, both stated openly, and its validation against higher-fidelity plasma models is named as the required next step. The filter is the certified floor beneath the learned and predictive controllers of this series.

Acknowledgments

Part of the Kronos Fusion Energy 2026 design series. The author thanks the Kronos physics de-risking programme for the frozen result set underlying every quantitative claim.

Reproducibility. Every headline quantity is a frozen anchor (gate KX-L1-A13 and companions) in the Kronos Physics De-Risking Register [43] (doi:10.5281/zenodo.22133057) and is regenerable from the deposited figure-generation script and the frozen anchors. This paper is archived at doi:10.5281/zenodo.22645716, and its official version is at <https://www.kronosfusionenergy.com/publications/control-barrier-safety-clamp.html>. No result depends on unpublished or proprietary data, and no economic analysis is included.

Scope — what this paper does not claim. The zero-escape guarantee holds for the reduced-order model and the declared disturbance bound; it is not a plant-level stability proof.

Data availability

The barrier model, the robust filter, the forward-invariance proof, the Monte-Carlo verification and the figure-generation script are archived with the deposit for this paper (doi:10.5281/zenodo.22645716), which cross-references the Kronos 2026 series including the breeder platform [34], the ideal-MHD stability analysis [35], and the AI/quantum control paper [42]. All quantitative values trace to the register [43] by the serial identifiers cited inline.

Competing interests

Provisional patent applications relating to aspects of this work have been filed.

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